

Intertidal Habitat Survey Point Defiance Shoreline

Ben Whitcomb ♦ GIS Field Data Collection ♦ June 6, 2026 ♦ 81 points • Owen Beach, Point Defiance

1 Field Data Points

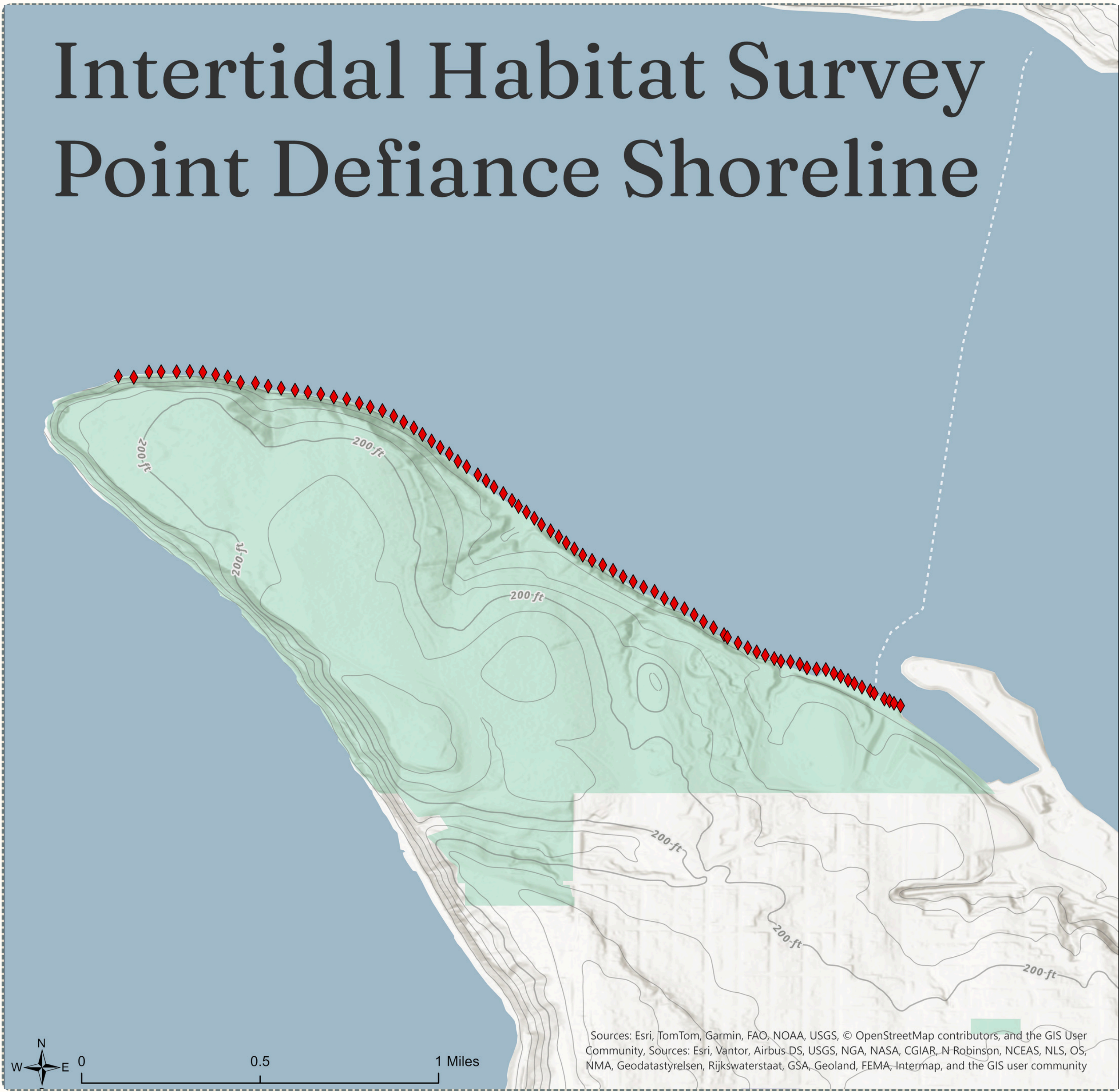


Fig 1. All 81 Field Maps observations along the Owen Beach transect, collected on foot at low tide.

81

Field points

35/22/24

High / Mid / Low zone

1.0–3.3

Tide range (ft)

2 Field Maps Workflow

Field Maps captured the quicklog data along the transect including primary and secondary substrate, intertidal zone, beach wrack cover, and shoreline modification at each point. The layer was built for speed on foot, with pick lists and a single photo field meant a full observation, the intention was to allow the transect to large areas of ground and focus biological surveys on important areas.

Four Arcade expressions added logic. A calculated point id that included date, beach name, and an auto generated point number, conditional visibility that hid the modification field outside the upper and mid intertidal, and conditional required rule forcing a photo on modified shoreline. Flagged points then linked to a detailed Survey123 form via a URL-encoded callback that auto-populated four fields so the deeper assessment never re-typed data already collected in field maps



Intertidal zone

Distribution of all 81 points



Intertidal zone was captured during survey as follows:

- High 3+ ft.
- Mid 1.5 - 3 ft.
- Low 1.5 ft. and below

Fig 2. Field Maps running in the field.

3 Survey123 Questions

At points determined to require biological surveying, an advanced Survey123 form was available. It opened with a site-and-conditions group whose point ID, beach, and substrate which were auto-populated from Field Maps, followed by surveyor, date and time, tide height, weather, and air temperature.

A biological-community group followed capturing both sessile and mobile species through multi-select lists with optional other species options, which would trigger a conditional single line text used to name additional species. These lists also included abundance ratings that appeared only when a species was recorded and an identification photo required at high density. A shoreline-modification group included armoring material, length, visible only when modification was present, and a riparian cover slider always present.

Three most-recorded species



8

Shore Crab

Hemigrapsus sp.



6

Little Brown Barnacle

Chthamalus dalli



3

Aggregating Anemone

A. elegantissima

4 Substrate & Community

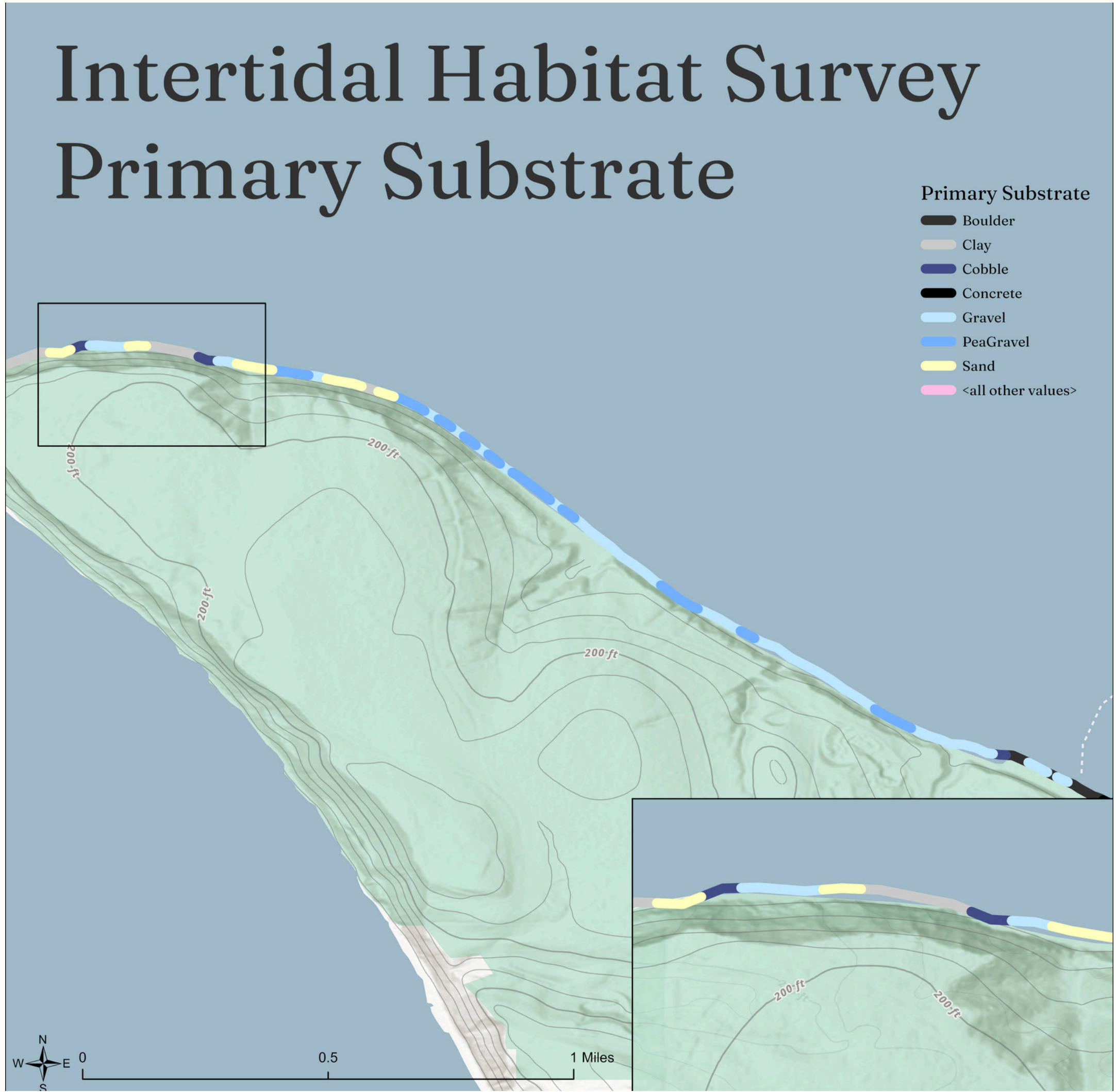
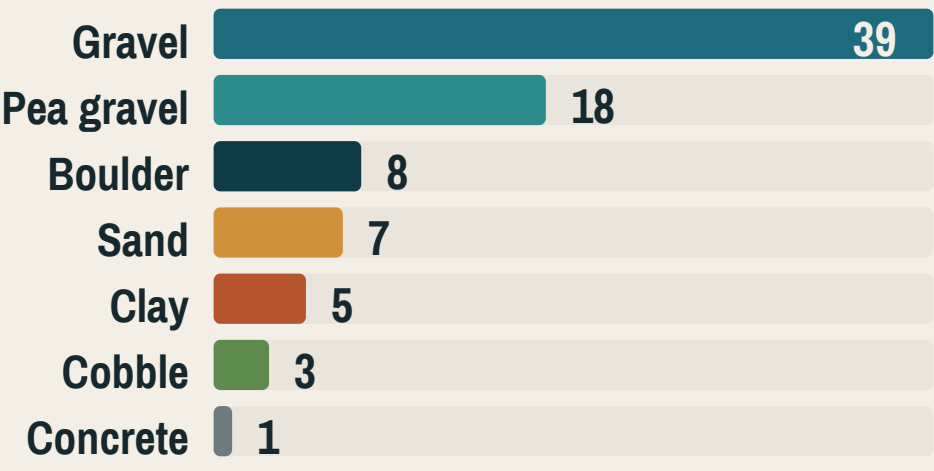
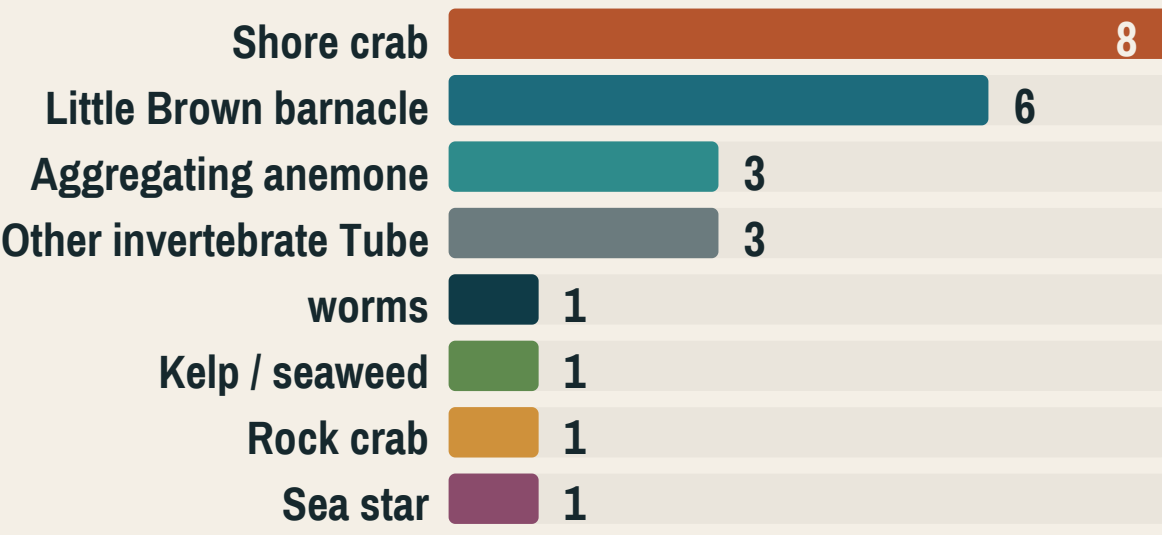


Fig 3. Owen Beach colored by primary substrate (ArcGIS Pro).

Primary substrate



Biological community



Gravel dominates the beach face (48% of points); shore crab and little brown barnacle are the most-detected taxa, richest in the less-modified low intertidal near the Point.